A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

# Transition metals and their chemistry

Copper sulfate is blue, potassium manganate(VII) purple, and iron catalyses the Haber process. These properties all come from one feature of the transition metals: an incomplete set of d electrons. This topic explains complex ions, their shapes and colours, the many oxidation states of vanadium and chromium, the reactions of metal aqua ions, and how transition metals act as catalysts.

## What makes a transition metal

A **transition metal** is a d-block element that forms one or more stable ions with an **incomplete d sub-shell**. Across the period from scandium to zinc, electrons are added to the 3d sub-shell, but when the atoms form ions they lose their **4s electrons first**.

| Element | Atom | Common ion | Ion configuration |
|---|---|---|---|
| Ti | [Ar] $3d^2$ $4s^2$ | $Ti^{3+}$ | [Ar] $3d^1$ |
| Cr | [Ar] $3d^5$ $4s^1$ | $Cr^{3+}$ | [Ar] $3d^3$ |
| Fe | [Ar] $3d^6$ $4s^2$ | $Fe^{2+}$, $Fe^{3+}$ | [Ar] $3d^6$, [Ar] $3d^5$ |
| Cu | [Ar] $3d^{10}$ $4s^1$ | $Cu^{2+}$ | [Ar] $3d^9$ |

- **Chromium and copper** have only one 4s electron: a half-filled ($3d^5$) or full ($3d^{10}$) d sub-shell is especially stable.
- **Scandium is not a transition metal**: its only ion, $Sc^{3+}$, is [Ar] with an *empty* 3d sub-shell.
- **Zinc is not a transition metal**: its only ion, $Zn^{2+}$, is [Ar] $3d^{10}$, a *full* d sub-shell.
- Because the 4s and 3d electrons are close in energy, successive ionisation energies rise gradually until all of them are removed, so several oxidation states are reachable.
- **Characteristic properties**: they form **complex ions**, form **coloured** compounds, show **variable oxidation states**, and act as **catalysts**.

## Complex ions and ligands

- **A complex ion** is a central metal ion surrounded by ligands bonded to it by dative (coordinate) bonds.
- **A ligand** is a molecule or ion with a lone pair of electrons that it donates to the metal ion to form a dative covalent bond, in which both shared electrons come from the same atom.
- **Coordination number**: the number of dative bonds to the central metal ion (not the number of ligands).

- **Monodentate** ligands form one dative bond: $H_2O$, $NH_3$, $Cl^-$, $OH^-$, $CN^-$.
- **Bidentate** ligands have two atoms that each donate a lone pair: ethane-1,2-diamine, $H_2NCH_2CH_2NH_2$ ("en"), and the ethanedioate ion, $C_2O_4^{2-}$. Three of them give a coordination number of 6, as in $[Cr(en)_3]^{3+}$ or $[Cr(C_2O_4)_3]^{3-}$.
- **Multidentate** ligands form several bonds: $EDTA^{4-}$ donates from two nitrogen atoms and four oxygen atoms, wrapping around a metal ion to form six bonds in a 1 : 1 complex. Haem is a multidentate ligand holding iron(II) in haemoglobin.
- **Charge of a complex** = charge of the metal ion + charges of the ligands: $Cu^{2+}$ with four $Cl^-$ gives $[CuCl_4]^{2-}$.

## Shapes of complex ions

| Shape | Coordination number | Bond angle | Examples |
|---|---|---|---|
| octahedral | 6 | 90° | $[Cu(H_2O)_6]^{2+}$, $[Ni(NH_3)_6]^{2+}$; most complexes with small ligands |
| tetrahedral | 4 | 109.5° | $[CuCl_4]^{2-}$, $[CoCl_4]^{2-}$: chloride ions are larger, so fewer fit around the metal |
| square planar | 4 | 90° | cisplatin, $[Pt(NH_3)_2Cl_2]$ |
| linear | 2 | 180° | $[Ag(NH_3)_2]^+$ (in Tollens' reagent), $[Ag(S_2O_3)_2]^{3-}$, $[Ag(CN)_2]^-$ |

## Cisplatin and haemoglobin

**Cisplatin** is a square planar platinum(II) complex with the two ammonia ligands next to each other (cis) and the two chloride ligands next to each other. Its isomer with them opposite each other, transplatin, has no anticancer action. In the body a chloride ligand is replaced by water, $[Pt(NH_3)_2Cl_2] + H_2O \rightarrow [Pt(NH_3)_2Cl(H_2O)]^+ + Cl^-$, and the platinum then forms dative bonds to nitrogen atoms on neighbouring bases in DNA, kinking the strand so the cancer cell cannot copy its DNA and divide. It also binds to the DNA of healthy cells, which causes side effects such as hair loss.

**Haemoglobin** carries oxygen as a ligand bonded to $Fe^{2+}$ in haem. Carbon monoxide is toxic because it forms a much stronger dative bond to the iron, replacing oxygen in a ligand exchange that is hard to reverse, so less oxygen can be carried.

# Why transition metal compounds are coloured

In an isolated ion the five 3d orbitals have the same energy. When ligands bond to the ion, their lone pairs repel the d electrons, and in an octahedral complex the d orbitals **split into two groups** at different energies, separated by

ΔE. A d electron can absorb a photon of visible light whose energy exactly matches ΔE and jump to the upper group:

$$\Delta E = hf$$

The light that is absorbed is removed from white light, and the colour we see is the **complementary colour** of the light that is transmitted or reflected. Hydrated copper(II) ions absorb in the orange-red region, so they look blue.

- **What changes the colour**: anything that changes ΔE: a different **ligand**, a different **coordination number** (and so shape), or a different **oxidation state** of the metal.
- **Colourless ions**: $Sc^{3+}$ ($3d^0$) has no d electrons to promote, and $Zn^{2+}$ and $Cu^+$ ($3d^{10}$) have no space in the upper d orbitals, so they absorb no visible light.

### Measuring concentration by colorimetry

The amount of light a coloured solution absorbs is proportional to the concentration of the coloured species. A colorimeter (spectrophotometer) uses a filter of the complementary colour, the colour the solution absorbs most strongly. If a complex is only pale, first add a ligand that forms a more intensely coloured complex. Make up solutions of known concentration, measure their absorbance, and plot a **calibration curve**; then measure the unknown and read its concentration from the graph.

## Variable oxidation states

The 4s and 3d electrons are similar in energy, so transition metals can lose different numbers of them. Some general trends: high oxidation states such as manganese(VII) in $MnO_4^-$ are oxidising agents; low ones such as $V^{2+}$ and $Fe^{2+}$ are reducing agents; and the +2 state becomes more stable relative to +3 across the period.

### Vanadium

| Oxidation state | Ion | Colour in solution |
|---|---|---|
| +5 | $VO_2^+$ (from ammonium vanadate(V), $NH_4VO_3$, in acid) | yellow |
| +4 | $VO^{2+}$ | blue |
| +3 | $V^{3+}$ | green |
| +2 | $V^{2+}$ | violet |

Adding zinc to acidified vanadate(V) reduces it step by step, through yellow, then green (a mixture of yellow and blue), blue, green and finally violet. Electrode potentials explain how far a reducing agent can go:

- $VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$, +1.00 V; $VO^{2+} + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O$, +0.34 V; $V^{3+} + e^- \rightleftharpoons V^{2+}$, −0.26 V.
- **Zinc** ($Zn^{2+}/Zn$ −0.76 V) is more negative than all three, so it reduces vanadium all the way to $V^{2+}$.
- **Tin** ($Sn^{2+}/Sn$ −0.14 V) is more negative than the first two but not the third, so it stops at $V^{3+}$.

## Chromium

- **Reducing dichromate(VI)**: zinc and hydrochloric acid reduce orange $Cr_2O_7^{2-}$ to green $Cr^{3+}$, then to blue $Cr^{2+}$: $Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$, then $2Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$. Air must be kept out (for example with a stream of hydrogen), because oxygen oxidises $Cr^{2+}$ straight back to $Cr^{3+}$.
- **Iron(II) is a weaker reducing agent**: $E^\ominus(Fe^{3+}/Fe^{2+})$ = +0.77 V lies between $E^\ominus(Cr_2O_7^{2-}/Cr^{3+})$ = +1.33 V and $E^\ominus(Cr^{3+}/Cr^{2+})$ = −0.41 V, so $Fe^{2+}$ reduces dichromate only as far as $Cr^{3+}$. $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$ is used in titrations (with an indicator, as the colour change is not sharp enough on its own).
- **Oxidising chromium(III)** is easier in alkaline solution, because removing electrons from a negative ion is easier. With excess NaOH, $Cr^{3+}$ forms green $[Cr(OH)_6]^{3-}$, which hydrogen peroxide oxidises to yellow chromate(VI): $2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 2OH^- + 8H_2O$.
- **Chromate and dichromate** are linked by an acid–base equilibrium, not a redox one (chromium is +6 in both): $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$. Adding acid turns yellow chromate orange; adding alkali removes $H^+$ and turns it back.

### Writing a half-equation in alkaline conditions

1. Start with the change: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-}$.
2. Balance oxygen with water and hydrogen with $H^+$, as if in acid: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-} + 2H_2O + 2H^+$.
3. Balance charge with electrons: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-} + 2H_2O + 2H^+ + 3e^-$.
4. Add $OH^-$ to both sides to turn each $H^+$ into water: $[Cr(OH)_6]^{3-} + 2OH^- \rightarrow CrO_4^{2-} + 4H_2O + 3e^-$.

## Reactions of metal aqua ions

A **Lewis acid** accepts an electron pair; a **Lewis base** donates one. In a complex the ligands are Lewis bases and the metal ion is a Lewis acid. Dissolved transition metal salts form hexaaqua ions, $[M(H_2O)_6]^{2+}$ or $[M(H_2O)_6]^{3+}$.

## Acidity of aqua ions

The metal ion pulls electron density from the oxygen of each water ligand, weakening its O–H bonds, so a proton can be lost to the solvent:

$$[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$$

3+ ions are smaller and more highly charged than 2+ ions, so they have a greater **charge density** and polarise the water ligands more strongly. Solutions of $M^{3+}$ ions are therefore noticeably acidic (around pH 2 to 3), while $M^{2+}$ solutions are only very weakly acidic. This hydrolysis is why iron(III) solutions look yellow-brown rather than pale violet.

## Deprotonation: hydroxide precipitates

Hydroxide ions and ammonia act as **Brønsted–Lowry bases**, removing protons from water ligands one at a time until a neutral complex forms, which precipitates:

- $[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(OH)_2(H_2O)_4] + 2H_2O$ (blue precipitate)
- $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(OH)_2(H_2O)_4] + 2NH_4^+$ (the same precipitate with a little ammonia)
- $[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow [Fe(OH)_3(H_2O)_3] + 3H_2O$ (orange-brown precipitate)

## Amphoteric hydroxides

The hydroxides of chromium(III) and zinc dissolve in excess sodium hydroxide, because further deprotonation gives a soluble negative ion; they also dissolve in acid. A hydroxide that reacts with both acids and alkalis is **amphoteric**:

- $[Cr(OH)_3(H_2O)_3] + 3OH^- \rightarrow [Cr(OH)_6]^{3-} + 3H_2O$ (dark green solution)
- $[Zn(OH)_2(H_2O)_4] + 2OH^- \rightarrow [Zn(OH)_4]^{2-} + 4H_2O$ (colourless solution)
- With acid: $[Cr(OH)_3(H_2O)_3] + 3H^+ \rightarrow [Cr(H_2O)_6]^{3+}$.

## Ligand exchange

- **With excess ammonia**, $NH_3$ replaces water (and hydroxide) ligands on copper, cobalt, nickel, chromium(III) and zinc, dissolving the precipitate. $NH_3$ and $H_2O$ are similar in size and both uncharged, so the coordination number stays 6 (exchange may be incomplete, as with copper): $[Cu(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2H_2O + 2OH^-$ (deep blue).
- **With concentrated HCl** (a high concentration of $Cl^-$), chloride replaces water. $Cl^-$ is larger than $H_2O$, so only four fit and the coordination number falls from 6 to 4, giving tetrahedral complexes: $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$ (yellow-green); $[Co(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CoCl_4]^{2-} + 6H_2O$ (pink to blue). Adding water reverses these.

## Stability of complexes and the chelate effect

Replacing monodentate ligands with bidentate or multidentate ones gives a much more stable complex. This **chelate effect** is explained by entropy:

- $[Cu(H_2O)_6]^{2+}$ + $EDTA^{4-}$ → $[Cu(EDTA)]^{2-}$ + $6H_2O$: 2 particles become 7.
- $[Co(NH_3)_6]^{2+}$ + 3en → $[Co(en)_3]^{2+}$ + $6NH_3$: 4 particles become 7.
- More particles means more ways to arrange the energy, so **$\Delta S_{system}$ is positive**. The same number and type of dative bonds are broken and made (M–O or M–N), so ΔH is close to zero and $\Delta S_{surroundings}$ is small. $\Delta S_{total}$ is therefore positive and the chelate complex is favoured.
- **Uses of EDTA**: it binds metal ions so tightly that it is used to remove toxic heavy-metal ions from rivers and from the body, and to soften water in shampoos by trapping $Ca^{2+}$.

### Working out formulae with silver nitrate

Chloride can be a ligand bonded to the metal or a free ion outside the complex. Silver nitrate precipitates only the **free** chloride ions as AgCl, so the mole ratio shows how many are outside. The three isomers of formula $CrCl_3·6H_2O$ behave differently:

| Complex | Mol AgCl per mol | Colour |
|---|---|---|
| $[Cr(H_2O)_6]Cl_3$ | 3 | violet |
| $[Cr(H_2O)_5Cl]Cl_2·H_2O$ | 2 | pale green |
| $[Cr(H_2O)_4Cl_2]Cl·2H_2O$ | 1 | dark green |

## Transition metals as catalysts

A catalyst speeds up a reaction by providing an alternative route with a lower activation energy, and is unchanged at the end. Transition metals are good catalysts because they can **change oxidation state** easily, and because they can use their 3d and 4s electrons to form weak bonds to reactants on their surface.

### Heterogeneous catalysis

A **heterogeneous** catalyst is in a different phase from the reactants, usually a solid with gases or liquids. The reaction happens on its surface:

1. Reactant molecules are **adsorbed** onto active sites on the surface, forming weak bonds to the metal.
2. Bonds within the reactants are weakened, and the molecules are held close together in a favourable orientation, so they react more easily.
3. The products are **desorbed** from the surface, freeing the sites.

- **Strength of adsorption** must be just right: tungsten adsorbs too strongly, so products are not released; silver too weakly, so too few reactant molecules stick. Nickel and platinum are in between and make good catalysts.
- **Surface area**: the catalyst is spread thinly on a support, such as a ceramic honeycomb, to give a large surface area from a small amount of expensive metal.
- **Advantages**: solid catalysts are easy to separate from the products and suit continuous processes.
- **Examples**: iron in the Haber process; vanadium(V) oxide in the Contact process, which changes oxidation state and changes back: $SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$, then $2V_2O_4 + O_2 \rightarrow 2V_2O_5$; platinum, palladium and rhodium in catalytic converters, which remove carbon monoxide, nitrogen monoxide and unburnt hydrocarbons, for example $2CO + 2NO \rightarrow 2CO_2 + N_2$.
- **Developing better catalysts**: ethanoic acid is now made from methanol and carbon monoxide, $CH_3OH + CO \rightarrow CH_3COOH$, with 100% atom economy, using an iridium–iodide catalyst. Successive catalysts have allowed milder conditions and better selectivity; such advances are published in peer-reviewed journals so other scientists can check them.

## Homogeneous catalysis

A **homogeneous** catalyst is in the same phase as the reactants. It works by forming an **intermediate**, usually by changing oxidation state and then changing back.

### Iron ions catalysing the reaction between peroxodisulfate and iodide

1. Overall: $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$. Uncatalysed it is slow, because two negative ions repel each other, so collisions with enough energy are rare.
2. Step 1: $2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$
3. Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$. The iron is back as $Fe^{2+}$, and $Fe^{3+}$ works equally well because the steps can happen in either order.
4. Both steps are between oppositely charged ions, so they are faster. This works because $E^\ominus(Fe^{3+}/Fe^{2+}) = +0.77$ V lies **between** $E^\ominus(I_2/I^-) = +0.54$ V and $E^\ominus(S_2O_8^{2-}/SO_4^{2-}) = +2.01$ V, so both steps are feasible.

*Electrode potentials show that a catalyst could work, but not that it will speed the reaction up: they say nothing about rates.*

## Autocatalysis

In **autocatalysis** a product catalyses the reaction. In the titration of ethanedioate with manganate(VII), $2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$, the reaction starts slowly (again two negative ions must collide), then speeds up as $Mn^{2+}$ builds up, because $Mn^{2+}$ provides a faster route:

- $4Mn^{2+} + MnO_4^- + 8H^+ \rightarrow 5Mn^{3+} + 4H_2O$
- $2Mn^{3+} + C_2O_4^{2-} \rightarrow 2Mn^{2+} + 2CO_2$
- The rate can be followed with a colorimeter measuring the purple colour, which does not disturb the reaction, or by titrating samples.

## Quick checks

**QUICK CHECK**

**What is the electron configuration of $Fe^{3+}$?**

- [Ar] $3d^3\ 4s^2$
- [Ar] $3d^5$ ✓
- [Ar] $3d^6$
- [Ar] $3d^4\ 4s^1$

Iron is [Ar] $3d^6\ 4s^2$; the two 4s electrons are lost first, then one 3d electron.

**QUICK CHECK**

**What is the coordination number of chromium in $[Cr(C_2O_4)_3]^{3-}$?**

- 3
- 4
- 6 ✓
- 9

Each ethanedioate is bidentate, forming two dative bonds: 3 × 2 = 6. The coordination number counts bonds, not ligands.

QUICK CHECK

**Why is a solution of $[Zn(H_2O)_6]^{2+}$ colourless?**

- zinc has no d electrons
- its 3d sub-shell is full, so no d–d electron transition is possible
- water is a weak ligand
- the complex is tetrahedral

$Zn^{2+}$ is $3d^{10}$. With every d orbital full, an electron cannot move to the upper set, so no visible light is absorbed.

QUICK CHECK

**Excess aqueous ammonia is added to aqueous copper(II) sulfate. What is seen?**

- a blue precipitate that stays
- a white precipitate
- a blue precipitate that dissolves to give a deep blue solution
- a yellow-green solution

A little ammonia deprotonates the aqua ion, giving blue $[Cu(OH)_2(H_2O)_4]$; excess ammonia then exchanges ligands to form deep blue $[Cu(NH_3)_4(H_2O)_2]^{2+}$.

**QUICK CHECK**

**Why are solutions of $[Fe(H_2O)_6]^{3+}$ more acidic than those of $[Fe(H_2O)_6]^{2+}$?**

- $Fe^{3+}$ has more water ligands
- $Fe^{3+}$ has a higher charge density, so it weakens the O–H bonds in its water ligands more
- $Fe^{3+}$ is a Brønsted–Lowry base
- $Fe^{2+}$ forms a precipitate with water

Both have six water ligands. The smaller, more highly charged $Fe^{3+}$ ion polarises them more, so they release $H^+$ more readily.

**QUICK CHECK**

**Which reaction is favoured mainly by a large increase in entropy?**

- $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CuCl_4]^{2-} + 6H_2O$
- $[Co(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 6H_2O$
- $[Ni(NH_3)_6]^{2+} + 3en \rightarrow [Ni(en)_3]^{2+} + 6NH_3$
- $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$

Four particles become seven when bidentate en replaces ammonia: the chelate effect. ΔH is close to zero because Ni–N bonds are broken and made.

**QUICK CHECK**

**Zinc and acid reduce vanadium(V) to vanadium(II), but tin only reduces it to vanadium(III). Why?**

- tin is a stronger reducing agent than zinc
- $V^{3+}$ is unstable
- [✓ $E^{\ominus}(Sn^{2+}/Sn)$ is more negative than $E^{\ominus}(VO^{2+}/V^{3+})$ but not more negative than $E^{\ominus}(V^{3+}/V^{2+})$]
- tin forms a complex with $V^{3+}$

$Sn^{2+}/Sn$ is −0.14 V. It can reduce couples with more positive $E^{\ominus}$ (+1.00 and +0.34 V) but not $V^{3+}/V^{2+}$ at −0.26 V. Zinc, at −0.76 V, can reduce all three.

**QUICK CHECK**

**In heterogeneous catalysis, why is silver a poor catalyst for many reactions?**

- it adsorbs reactants too strongly
- [✓ it adsorbs reactants too weakly, so few stay on the surface long enough to react]
- it is in the same phase as the reactants
- it changes oxidation state too easily

Too weak adsorption (silver) and too strong adsorption (tungsten) both make poor catalysts. Nickel and platinum adsorb with intermediate strength.